Dream → Chairs - approaches & design paths

This is a toolkit of interchangeable ideas. We've prototyped a set of building blocks and several ways to connect them. At each stage there are multiple paths; pick and combine the ones that fit your design intent.

One connective idea

don't map dream → chair directly - map both onto body posture and use that as the shared coordinate. A chair proposes *how to hold a body*; a dream stages *a body in space*. The common variable is posture. It's a great organizing principle **if** you want the chair to express the dream's bodily feeling.

The building blocks that exist and work

- **Text → 3D form** - Shap-E (meshes) ✓ and Point-E (point clouds) ✓.
- **A shared 3D↔text space** - OpenShape encodes a form into the *same space as words*, so forms and text are directly comparable with no rendering ✓.
- **Descriptor vocabularies** - design families + a large visual dictionary ✓.
- **Clustering + characterization** of a whole corpus ✓.
- **Posture axes** (the dream↔body↔chair bridge) ✓.

Menu of design paths

1 · What is the unit of design?

- **Individual dream → one chair** - an idiosyncratic, singular object. ✓
- **A collection → families → an archetype chair** per family (decode the cluster centroid; let within-family variance become deformation: spread→asymmetry, instability→tilt). ◦ The question becomes 'how do we define a family?' and 'in what dimensional space?' (shape, text, posture, symbol).

2 · How to find "families" of dreams?

Each clusters something different and yields different families. **They can be combined.**

- **Cluster the 3D shapes themselves** - group the generated point clouds / OpenShape embeddings to discover *families of forms* emerging from the data, then let those emergent shape-types inform the final results. Data-driven and surprising. ✓
- **Cluster the dream texts** - group by what the reports *say* (text embeddings); families by narrative/semantic content rather than form. ◦
- **Cluster by posture** - group by the 6 body-axis coordinates (derived from text or shape); families by *bodily attitude* (all the "reclined/cocooned" dreams together). ◦
- **Group by Lakota symbol** - assign each dream its top symbol and cluster by symbol - a culturally grounded grouping. Needs Knowledge-Holder descriptors to be real. ◦ / ✗

- **Hybrid** - e.g. symbol first, then shape-family within each symbol; or posture within a shape-family.

3 · How to *characterize / name* a family

- **Design-descriptor families** - volume, form language, symmetry, edge, proportion, surface, posture - via contrastive signatures (what a cluster has *more* of than average). ✓
- **Emergent visual descriptors** - a large neutral dictionary, data-driven. ✓
- **LLM labels** - generate a short text label for each cluster (e.g. "tall spire", "cocooned pod"). ✓
- **Lakota symbols + their qualities** - name families by symbol and the spatial qualities attached to it (cultural grounding). ◦ / ✗

4 · Linking dream → chair (the body)

- **Posture from the dream text** (LLM rubric for nuanced reports, or zero-shot CLIP for short ones) → chair parameters. ✓
- **Posture from the 3D shape** (project the form onto the body axes) → chair parameters. ✓
- **Generate-and-select** - generate several chairs from the posture, keep the one whose body best matches the dream. ✓
- **Archetype chair** - the chair of all "falling" dreams, from a cluster centroid. ◦

5 · Measuring dream ↔ form alignment (faithfulness)

Does text→3D actually preserve a dream's structure? Embed the dream and its generated form in the shared space, project **both** onto the same dimensions (symbols, posture, design qualities), and compare the two profiles. Two uses:

- **Which properties survive translation** - *per-dimension*, across the corpus: which axes track between text and form (e.g. enclosure/posture preserved r≈0.4, symbolic identity largely lost r≈0.1)? ✓
- **Pick the strongest individual dreams** - *per-dream*: score how well each form kept its own dream's signature, then keep the top-aligned dreams for the one-dream → one-chair route. ✓
  Method note: standardize within modality (text↔text and form↔text cosines differ in scale) and compare against a shuffled baseline, so "preserved" means above chance. (`scripts/alignment.py`)

Open knobs

These are where design + phenomenology judgment should drive the system.

- **A · Posture axes** (`data/descriptors/body_posture_axes.json`) - are the 6 axes, their poles, and the chair-parameter mappings right? Check the axis-extreme sheet; each hand-score ~10 dreams to compare against the auto-extraction.
- **B · Which substrate defines a family** - shapes, text, posture, or symbols ?
- **C · Design descriptors that name clusters** (`data/descriptors/design_families.json`, `visual_descriptors.json`) - right vocabularies? Missing categories (seat geometry, leg structure, back profile)? Edit and re-cluster easily.
